

JEE MAIN REVISION MASTER: SALT ANALYSIS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 43 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Salt Analysis - Core Application	43 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Salt Analysis - Core Application (Very High Yield — 43 PYQs)

Core Principle: Key problem pattern in Salt Analysis.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Among Fe^{3+} , Pb^{2+} , Cu^{2+} and Mn^{2+} , identify the one that gets precipitated out while passing H_2S in presence of NH_4OH ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider three metal chlorides x,y and z, where x is water soluble at room temperature, y is sparingly soluble in water at room temperature..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Among Fe^{3+} , Pb^{2+} , Cu^{2+} and Mn^{2+} , identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

- (A) +3
- (B) +4
- (C) +2
- (D) +7

Correct Answer: None

Solution Analysis: While passing H_2S in the presence of NH_4OH , the medium becomes basic. In basic medium, the concentration of S^{2-} ions increases, so sulphides of some metal ions get precipitated. Now let us check the given ions:
 Fe^{3+} : does not precipitate here as sulphide in this group
 Pb^{2+} : belongs to earlier group and is precipitated as chloride, not here
 Cu^{2+} : precipitates as CuS
 Mn^{2+} : can precipitate as MnS in ammoniacal medium But in classical qualitative analysis, passing H_2S in presence of NH_4OH is used for precipitating group IV cations like Mn^{2+}